



May 2025

(For AM)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

10. Which of the following are necessary conditions for a deadlock to occur?
- A) Mutual exclusion, paging, demand paging, thrashing
B) Deadlock prevention, deadlock detection, deadlock recovery
C) Contiguous memory allocation, demand paging, swapping
D) Mutual exclusion, hold and wait, no preemption, circular wait
11. The circular wait condition for deadlock implies that
- A) There exists a set of waiting processes such that each process is waiting for a resource held by the next process in the set
B) Every process is waiting for a single resource that is being used by another process
C) Processes do not wait for resources
D) Each process executes in isolation without requesting resources
12. What is the main drawback of deadlock avoidance techniques?
- A) They are too complex to implement
B) They cannot detect deadlocks
C) They require frequent swapping of processes
D) They require prior knowledge of resource requests
13. What is the primary purpose of paging in memory management?
- A) To allow programs to run faster
B) To allow programs to run faster
C) To avoid fragmentation and provide efficient memory allocation
D) To increase CPU speed
14. Which of the following best describes a page table?
- A) A table that stores physical addresses of processes
B) A table that stores logical-to-physical address mappings
C) A list of all the pages in the hard disk
D) A cache memory management table
15. Which type of fragmentation is eliminated by paging?
- A) Internal fragmentation
B) Both internal and external fragmentation
C) External fragmentation
D) Does not eliminate fragmentation
16. Copy-on-Write (COW) is used in virtual memory to
- A) Copy data from disk to RAM
B) Avoid memory wastage by sharing pages until modification
C) Improve CPU scheduling
D) Reduce page size
17. What is the main advantage of hierarchical directory structures?
- A) Faster file access
B) Reduces disk usage
C) Eliminates file fragmentation
D) Better file organization and grouping
18. Which file system structure allows a single file to appear in multiple directories without duplicating its contents?
- A) Single-Level Directory
B) Hierarchical Directory
C) Acyclic-Graph Directory
D) Two-Level Directory
19. Which file allocation method uses an index to keep track of all file blocks?
- A) Contiguous Allocation
B) Linked Allocation
C) Indexed Allocation
D) Direct Allocation
20. Which of the following best describes a Type 1 hypervisor?
- A) Runs on a host operating system
B) Requires another hypervisor to function
C) Only supports full virtualization
D) Runs directly on the hardware without a host OS

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following operating system architecture in detail: (i) Monolithic Systems (ii) Layered Approach	8	L*2	1	1,2

- b) Consider the following set of processes with their respective arrival times and burst times. Using the Round Robin (RR) scheduling algorithm with a time quantum of 2 ms, determine the following for each processes: gantt chart, completion time, turn around time and waiting time.

Process	Arrival Time	Burst Time
P1	0	5
P2	2	8
P3	4	3
P4	6	6
P5	8	7
P6	10	4

2. a) Depict with a diagram and explain the concept of context switching between two processes. How do system calls enable the transition between user and kernel modes? Explain with a neat diagram.

8 L3 2 1,2

- b) (i) Explain how Test-and-Set lock instructions can be used to solve the critical section problem. Provide a suitable example to illustrate your solution.

8 L2 1 1,2

(ii) Illustrate the working of a counting semaphore with five processes (P1, P2, P3, P4, P5), where only three processes can enter the critical section at a time. Given that P1, P2, and P3 enter at 1ms, 2ms, and 3ms respectively, while P4 and P5 must wait until a process completes execution (P1 exits at 6ms and P2 exits at 7ms), explain the entire process with proper synchronization using the semaphore operations Down() and Up(). Provide a step-by-step illustration of the execution.

Struct Semaphore

```
{ int value;
  Queue type L;
}
Down (Semaphore S)
{
  S.value = S.value -1;
  if(S.value<0)
  {
    put process in L;
    sleep();
  }
  else
    return;
}
```

Up(Semaphore S)

```
{
  S.value = S.value +1;
  if(S.value<=0)
  {
    Select a process from L;
    wakeup();
  }
}
```

8 L3 2 1,2

3. a) What is an operating system? Describe the structure of an operating system and explain each component in detail.

8 L2 1 1,2

- b) Explain the process lifecycle with a diagram. Discuss the classical IPC problems: Dining Philosophers Problem. Explain their challenges and possible solutions using semaphore.

8 L3 2 1,2

Unit – II

4. a) Explain the Safety Algorithm and Resource Request Algorithm in Detail. 8 L2 3 1,3
- b) Explain the Buddy system allocation in an operating system. A system has 4 page frames and the following page reference string: 7, 0, 1, 2, 0, 3, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Apply the LRU page replacement algorithm and compute the total number of page faults. 8 L3 4 1,3
5. a) Consider a system with five threads T0 through T4 and four resource types A, B, C and D. Suppose that the following snapshot represents the current state of the system:

	<u>Allocation</u>	<u>Max</u>	<u>Available</u>
	<u>A B C D</u>	<u>A B C D</u>	<u>A B C D</u>
T ₀	0 0 1 2	0 0 1 2	1 5 2 0
T ₁	1 0 0 0	1 7 5 0	
T ₂	1 3 5 4	2 3 5 6	
T ₃	0 6 3 2	0 6 5 2	
T ₄	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

- a. What is the content of the matrix Need?
- b. Is the system in a safe state?
- c. If a request from thread T1 arrives for (0,4,2,0), can the request be granted immediately? 8 L3 3 1,3
- b) What is demand paging? Explain the steps involved when a page fault occurs in a system using demand paging. 8 L2 4 1,3
6. a) Describe the four necessary conditions for a deadlock to occur. Illustrate with a real-world example. 8 L3 3 1,3
- b) Consider a memory block list of size: 100KB, 500 KB, 200 KB, 300 KB, 600 KB. Processes arrive with memory requests: 212 KB, 417 KB, 112 KB, 426 KB. Apply Best Fit and First Fit allocation strategies and calculate: (i) Memory allocation table for each strategy (ii) Internal fragmentation (iii) External fragmentation. 8 L3 4 1,3

Unit – III

7. a) Explain how tree-structured directories and Acyclic-graph directories organizes files and directories. 8 L2 5 1,3
- b) Explain the continuous and indexed allocation methods in detail. 8 L2 5 1,3
8. a) Explain the difference between Type 1 and Type 2 hypervisors. Discuss their advantages and use cases with suitable examples. 8 L2 5 1,3
- b) Discuss virtual machine migration. What are the key challenges in live migration, and how does it benefit cloud computing? 8 L2 5 1,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
